IP ADDRESSING AND SUBNETTING

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Class	IP Address Range	Used For	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large networks	Google or Microsoft company network
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	University or college network
Class C	192.0.0.0 – 223.255.255.255	Small networks	Home Wi-Fi router (192.168.1.1)
Class D	224.0.0.0 – 239.255.255.255	Multicast	Live YouTube streaming or IPTV
Class E	240.0.0.0 – 255.255.255.255	Experimental/Research	Network research and testing

2. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

- Your IP address 192.168.1.300 is invalid because each octet in an IPv4 address can only have a value from 0 to 255. The number 300 is outside the valid range, so the device cannot connect to the network.
- For the 192.168.1.0/24 subnet, a valid host IP address would be between 192.168.1.1 and 192.168.1.254 (assuming .0 is the network address and .255 is the broadcast address).

**3. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.**

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IP address :- 192.168.10.45

Subnet Mask :- 255.255.255.0

According to subnet Mask the

CIDR value: /24

So, 192.168.10.0 [Network address]
192.168.10.255 [Broadcast address]

* Host

$2^n - 2$

$= 2^8 - 2$

$= 256 - 2$

$= 254$

* IP table

$\boxed{192.168.10.0}$ — Network Address

192.168.10.1

192.168.10.253

$\boxed{192.168.10.254}$ — Broadcast Address.

4. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Group	Network Address	Subnet Mask	CIDR	Host Range
Delivery Partners	10.0.0.0	255.255.255.192	/26	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64	255.255.255.192	/26	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128	255.255.255.192	/26	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192	255.255.255.192	/26	10.0.0.193 – 10.0.0.254

**5 . You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.**

IP Address :- 172.16.0.0/16

Original Subnet mask :- 255.255.0.0

Required subnets :- 20

→ $2^n \geq$ Required subnet

$2^4 = 16$ ×

$2^5 = 32$ ✓

So, we need 20 subnet so we need to increase network range.

Need to take 5 bits from next octet.

$\underbrace{11111111.11111111.11111}_{21\,N}\underbrace{000.00000000}_{11\,Host}$.

255 . 255 . 248 . 0

→ Calculating Host per subnet.

$2^H - 2$

$= 2^{11} - 2$

$= 2048 - 2$

$= 2046$ Host per subnets